

# Python Environment Setup Guide (Windows)

## Install PyEnv for Windows

### Install Python versions

### Set a global Python version

### Configure a project-specific Python version

### Create and activate a virtual environment

#### Step 1: Install PyEnv for Windows

Open PowerShell as Administrator

Click Start

Search PowerShell

Right-click Windows PowerShell

Select Run as Administrator

#### Install PyEnv

PowerShell

```
Invoke-WebRequest -UseBasicParsing -Uri "https://pyenv-win.github.io/pyenv-win/install.ps1" -OutFile  
"./install.ps1"  
& "./install.ps1"
```

#### If Script Execution is blocked

PowerShell

```
Set-ExecutionPolicy RemoteSigned -Scope CurrentUser
```

Then run install command again.

#### Step 2: Restart Terminal

Close all PowerShell windows

Open a new PowerShell window

#### ✔ Verify installation

PowerShell

```
pyenv --version
```

#### Step 3: Install Python Versions

Install Python 3.12.10

PowerShell

```
pyenv install 3.12.10
```

Optional versions

PowerShell

```
pyenv install 3.10.11
```

```
pyenv install 3.11.9
```

#### ✔ Check installed versions

PowerShell

```
pyenv versions
```

#### Step 4: Set Global Python Version

Set default Python

PowerShell

```
pyenv global 3.12.10
```

#### ✔ Verify

PowerShell

```
python --version
```

## Step 5: Configure Project Python Version

Go to project folder

PowerShell

```
cd D:\Projects\MyProject
```

Set project-specific Python

PowerShell

```
pyenv local 3.12.10
```

✓ **Verify**

PowerShell

```
python --version
```

## Step 6: Create Virtual Environment

Create .venv

PowerShell

```
python -m venv .venv
```

## Step 7: Activate Virtual Environment

PowerShell

```
.\venv\Scripts\Activate
```

✓ **You will see:**

```
(.venv)
```

## Step 8: Verify Virtual Environment

Check Python path

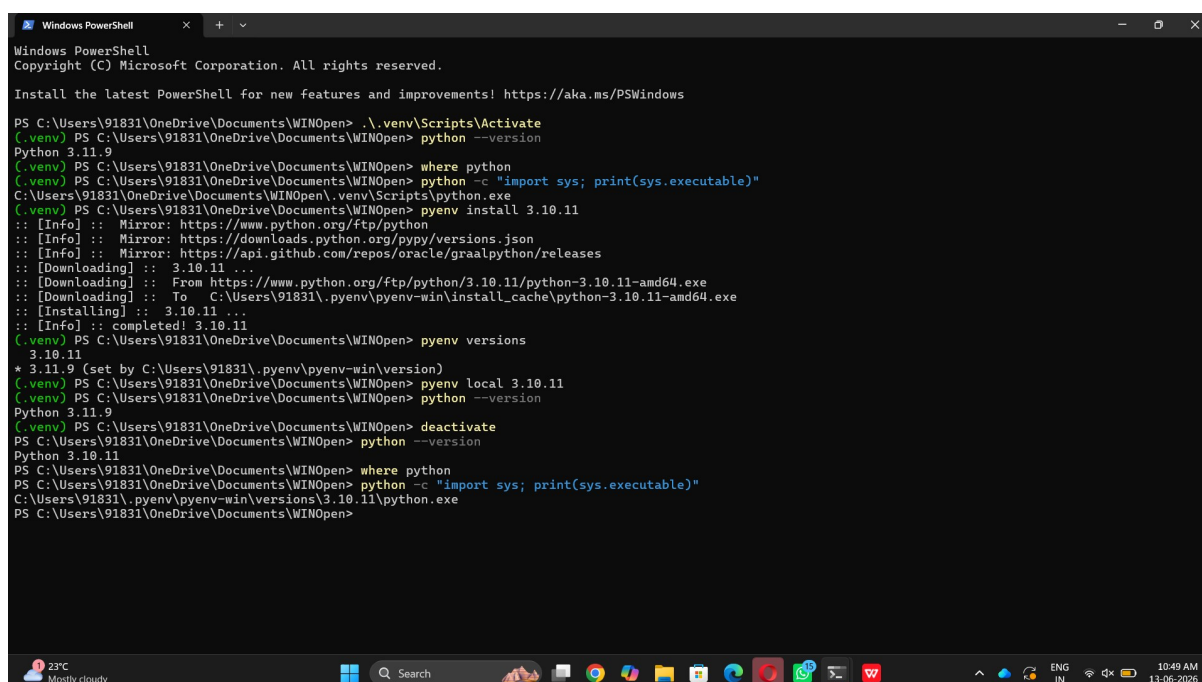
PowerShell

where python

Confirm executable

PowerShell

```
python -c "import sys; print(sys.executable)"
```



```
Windows PowerShell
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Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\91831\OneDrive\Documents\WINOpen> .\venv\Scripts\Activate
(.venv) PS C:\Users\91831\OneDrive\Documents\WINOpen> python --version
Python 3.11.9
(.venv) PS C:\Users\91831\OneDrive\Documents\WINOpen> where python
(.venv) PS C:\Users\91831\OneDrive\Documents\WINOpen> python -c "import sys; print(sys.executable)"
C:\Users\91831\OneDrive\Documents\WINOpen\.venv\Scripts\python.exe
(.venv) PS C:\Users\91831\OneDrive\Documents\WINOpen> pyenv install 3.10.11
:: [Info] :: Mirror: https://www.python.org/ftp/python
:: [Info] :: Mirror: https://downloads.python.org/pypy/versions.json
:: [Info] :: Mirror: https://api.github.com/repos/oracle/graalpython/releases
:: [Downloading] :: 3.10.11 ...
:: [Downloading] :: From https://www.python.org/ftp/python/3.10.11/python-3.10.11-amd64.exe
:: [Downloading] :: To C:\Users\91831\.pyenv\pyenv-win\install_cache\python-3.10.11-amd64.exe
:: [Installing] :: 3.10.11 ...
:: [Info] :: completed! 3.10.11
(.venv) PS C:\Users\91831\OneDrive\Documents\WINOpen> pyenv versions
3.10.11
* 3.11.9 (set by C:\Users\91831\.pyenv\pyenv-win\version)
(.venv) PS C:\Users\91831\OneDrive\Documents\WINOpen> pyenv local 3.10.11
(.venv) PS C:\Users\91831\OneDrive\Documents\WINOpen> python --version
Python 3.11.9
(.venv) PS C:\Users\91831\OneDrive\Documents\WINOpen> deactivate
PS C:\Users\91831\OneDrive\Documents\WINOpen> python --version
Python 3.10.11
PS C:\Users\91831\OneDrive\Documents\WINOpen> where python
PS C:\Users\91831\OneDrive\Documents\WINOpen> python -c "import sys; print(sys.executable)"
C:\Users\91831\.pyenv\pyenv-win\versions\3.10.11\python.exe
PS C:\Users\91831\OneDrive\Documents\WINOpen>
```